Review the assignment due date

Final Project

Team Number: T16

Team Name: Sewing Machine

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GitHub Repository URL: https://github.com/upenn-embedded/final-project-s26-t16-new

GitHub Pages Website URL: [for final submission]*

Final Project Proposal

1. Abstract

Our final project is a smart sewing machine with a wireless foot pedal and built-in thread usage tracking. The wireless pedal will use a pressure sensor to detect how hard the user is pressing and send that input to the sewing machine, which will adjust motor speed in real time. A second part of the system will track spool or shaft rotation to estimate how much thread remains and display that information to the user on an OLED or other indicator. Overall, the project combines wireless communication, sensor input, motor control, and user feedback into one integrated embedded system.

2. Motivation

Conventional sewing machines usually use wired pedals and provide limited real-time information about machine operation. This creates usability limitations, including restricted pedal placement and difficult portability. Furthermore, there is no clear indication of the remaining thread or machine state. This is especially annoying when sewing complex pieces; users have to guess and check when to replace bobbin and spool thread and precision is limited by eyeballing. Our project addresses this by adding a wireless pressure-sensitive pedal and a sensing system that estimates thread usage during operation. This project is interesting because it requires real-time interaction between mechanical and embedded subsystems. The device must sense pedal pressure, filter and transmit the input, adjust motor speed, count machine rotations, and provide output through a display or indicators. The intended purpose is to build a more flexible and informative sewing machine system that improves user control while demonstrating strong embedded system integration. It is also just cool because some members in our group love to sew and the chain locking mechanism is a marvel in itself, so it could be a cool mechanism to figure out.

3. System Block Diagram

alt text

We will need to 3D print the sewing machine components. Further, for the body and internal support of the machine, we will need power tools to cut it out of wood or acrylic.

4. Design Sketches

alt text

5. Software Requirements Specification (SRS)

Pedal Pressure to Speed Mapping
The pedal MCU reads pressure and maps it to motor speed (higher pressure → higher speed).
Validation: Apply ≥5 pressure levels and verify speed increases monotonically.

Wireless Command Transmission
The pedal MCU sends speed commands wirelessly to the sewing machine MCU.
Validation: Perform ≥20 trials with ≥95% successful reception and response.

Real-Time Motor Control The machine MCU adjusts motor speed based on pedal input.
Validation: Test ≥5 inputs and verify speed settles within 2 s.

Rotation Counting
The MCU tracks rotations using an encoder, Hall, or optical sensor.
Validation: Compare measured vs. actual rotations with ≤5% error.

Thread Estimation
Remaining thread is estimated from rotation count and spool geometry.
Validation: Compare estimate vs. expected value with ≤10% error.

User Feedback
System provides status via LED/OLED (power, thread, state).
Validation: Verify LED power indication and display updates during operation.

Chain Stitch Mechanism
The mechanism produces a continuous chain stitch.
Validation: Test on fabric and verify stitch remains continuous.

Power Transmission
Motor drives stitching mechanism without slip or stall.
Validation: Test across speed range and verify consistent motion.

Frame Stability
Structure remains stable during operation.
Validation: Run for several minutes and check for vibration or loosening.

Pedal Usability
Pedal supports repeated presses and returns to neutral.
Validation: Perform ≥50 cycles and verify consistent return and integrity.

Spool Support
Spool rotates smoothly and remains mounted.
Validation: Run repeated tests and verify no jamming or detachment. |

6. Hardware Requirements Specification (HRS)

Wireless Foot Pedal Hardware
The system includes a portable foot pedal with an embedded pressure sensor and microcontroller to detect user input and transmit speed commands wirelessly.
Validation: Apply repeated low, medium, and high pressure inputs and verify that the pedal produces distinct sensor readings and successfully transmits commands.

Dual-MCU Architecture
The system uses two separate microcontrollers: one in the wireless pedal for pressure sensing and transmission, and one on the sewing machine for motor control, display output, and spool/thread monitoring.
Validation: Test each MCU separately, then validate the full system by confirming that pedal input is received and acted on by the machine-side MCU.

Motor Drive Hardware
The sewing machine includes a motor and motor driver capable of rotating the stitching mechanism at variable speeds based on pedal input.
Validation: Operate the motor at multiple commanded speeds and verify continuous rotation and response to speed changes without stalling under normal operation.

Rotation Sensing Hardware
The machine includes a sensor, such as an encoder, Hall-effect sensor, or optical sensor, to detect spool or shaft rotation for thread-usage estimation.
Validation: Rotate the shaft or spool through a known number of turns and verify that the sensor output matches the actual number of rotations.

User Feedback Hardware
The system includes at least one output device to indicate machine status, such as an OLED display or LED indicators.
Validation: Power on the system and confirm that the display or indicators activate correctly and update during operation.

Power System
The device includes a power source and regulation hardware capable of safely powering both microcontrollers, the wireless pedal, sensors, display, and motor-driving circuitry.
Validation: Run the full system under normal operation and verify that all components remain powered and functional without unintended resets or brownouts.

Pressure Sensor Resolution
The foot pedal pressure sensor provides enough range to distinguish at least three input levels: low, medium, and high pressure.
Validation: Record sensor outputs at different applied pressures and confirm that the three levels are clearly distinguishable.

7. Bill of Materials (BOM)

BOM

First, the source of our compute/signal processing for this project is the ATmega328PB XPlained mini board. This is ideal because it provides sufficient compute for processing of simple sensor data, is cheap and highly flexible when tuned with baremetal code. To facilitate wireless communication between two of these boards, a Feather ESP32 Wi-Fi module will be connected to each MCU. This is preferred because the Feathers provide bluetooth capabilities that are ideal for short distance, light-weight communication. They also provide access to wifi which gives us the flexibility of incorporating data storage, and IOT with this system without requiring significant changes in storage. For our power source, we will utilize 3 sources in our testing phase. A 5V power bank will be utilized to power our foot pedal. A power bank is preferred because they have built in power regulation and are small and portable which is the entire purpose of having a detached foot pedal. 5V adapters plugged into the walls will be used to power the peripherals on the sewing machine side. For the motor, a separate 12V supporting 2A power adapter plugged into the wall will be used. We want to separate the power supply of the motor from the peripherals because the motor can cause significant current/voltage spikes. Our pedal must be able to distinguish between multiple states which we do using a pressure sensor. We chose the FSR 402 because it has a sensing range of 0.3N to 150N which is the relative range that makes the pedal easy to control and not require too much force. The sensor is also durable supporting 10 million + actuations and has mm thickness. For our motor, we choose the FIT0493 motor because it is relatively inexpensive, contains a built in encoder, supports PWM and should have more than enough torque at 12kg/cm. It takes 12V and draws a max current of 1.65A. Our motor driver supports this by being able to provide up to 3A. This is all supported by our 12V adapter. We also require an LCU to display information to the user. In all honesty, the specs for this are unimportant for the operation of the device as long as SPI communication is supported. We also use a hall-effect latch for spool rotation tracking rather than an optical sensor because sewing might produce significant debris and a magnetic sensor handles that well. Our chosen sensor takes in 5V and only requires 5mA to operate which matches well within the rest of the system.

8. Final Demo Goals

Will it be strapped to a person, mounted on a bicycle, or require outdoor space?

- It is a standalone machine with a portable foot pedal and connected to a battery, so no need to be strapped to anything or anyone.

Think of any physical, temporal, and other constraints that could affect your planning.

- Some constraints that can affect planning is that a sewing machine isn't something that we can fit in a bag; it is quite big, so storage could be a constraint as well as methods to transport the parts back and forth.

9. Sprint Planning

How will you plan your sprint milestones?

- We will plan our sprint milestones as semi-long term goals. Focus on 1-2 big goals max and separate into 2 person groups to complete mini parts that work up to completing the overarching goal. For example if the unit time is 1 week, we will have the goal we want completed by the end of the week

defined clearly. After defining this goal, define the sub goals that will be completed to complete the overall goal. Then, block out specific time blocks throughout the week to fulfill each goal. Finally, leave a few hours in case some tasks take longer than anticipated.

How will you distribute the work within your team?

- Everyone will be present when planning the sprints. We will just discuss which work we want to do and split accordingly. Nothing will be assigned without approval and we will discuss during in person sprint planning and over text for unanticipated tasks that come up.

This is the end of the Project Proposal section. The remaining sections will be filled out based on the milestone schedule.

Sprint Review #1

Last week's progress

This past week, we finalized BOMed (and order), worked on an intial physical CAD (and sent out to print) and wrote an inital version of the embedded code for the foot pedal system.

In terms of the BOM, the biggest change we made compared to the inital write up was we switchd the force sensor to a different model with a larger surface area for ease of integration and use. We also realized the power adapter we were planning on using does not source enough current to account for spikes in the motor so we made changes to this.

ESE3500 S26 Final Project BOM

	A	B	C	D	E	F	G	H	I	J
1	**Team Number**	16								
2	**Team Name**	Team 16								
3		**Name**	**Email**	**GitHub Handle**						
4	**Team Member # 1**	Chris Chen	chrisc05@seas.upenn.edu	chrishchen29						
5	**Team Member # 2**	Ivy Jiang	ivyjiang@seas.upenn.edu	ivyjiangg						
6	**Team Member # 3**	Evelyn Li	eli22@seas.upenn.edu	2024eli				**Total Cost**		
7	**Team Member # 4**	Daniel Lin	dl1n@seas.upenn.edu	d-l1n						
8	**Row of totals**							**$ 87.03**		
9	**Part Role**	**Part Description**	**Manufacturer**	**Manufacturer part number (MPN)**	**Interface to MCU**	**MCU Pins Assigned**	**Distributor**	**Cost per device**	**URL**	**Comments**
10	Processor	Main Processor	Microchip	ATmega328PB Xplained Mini	N/A	N/A		$ -		Already included in kit
11	Processor	ESP-32 Wi-Fi Module	Adafruit	5438	UART	PD0, PD1		$ -		Find in Detkin (x2)
12	Sensor (Input)	foot pedal pressure sensor	Interlink electronics	34-00153	ADC	PC4, PC5	DigiKey	$ 8.90	https://www.digikey.co	
13	Actuator (Output)	Motor with encoder	DFRobot	FIT0493	GPIO	PD2	DigiKey	$ 29.00	https://www.digikey.co	
14	Actuator (Output)	Motor Driver	Pololu	DRV8874	PWM	PD3, PD4	Pololu	$ 12.00	https://www.pololu.co	
15	Actuator (Output)	LCD Screen	Adafruit	938	I2C	PC4, PC5		$ -		Find in Detkin
16	Sensor (Input)	Hall Sensor	Melexis Technologies NV	US1881LUA-AAA-000-SP	GPIO	PD7	DigiKey	$ 0.60	https://www.digikey.co	
17	Miscellaneous	5V Power Bank	Adafruit	4288	N/A	N/A	Adafruit	-	https://www.adafruit.c	probably don't need to buy this, we can get one in detkin/team members have some we can use for testing
18	Miscellaneous	5V USB Wall Adapter	Adafruit	1994	N/A	N/A	Adafruit	-	https://www.adafruit.c	find in detkin
19	Miscellaneous	12V Motor Supply	StarTech	SVA12M2NEUA	N/A	N/A	DigiKey	$ 29.54	https://www.digikey.co	motor power supply
20		Barrel Power Cable	Tensility	10-03609			DigiKey	$ 6.99	https://www.digikey.co	
21										
22										
23		ORDER PARTS BELOW:								
24		Force-sensitive resistor	https://www.digikey.com/en/p	$10.08						
25		Motor with encoder	https://www.digikey.com/en/p	$29.00						
26		Motor Driver	https://www.pololu.com/prod	$11.94						
27		Motor supply adapter	https://www.digikey.com/en/p	$6.99						
28		12V Motor Supply	https://www.mouser.com/Pro	$22.75						
29				**$80.76**						

In terms code for the foot pedal, we have written and tested the functionality for measuring and reading the force from the sensor. This is the core of the control system.

upenn-embedded / final-project-s26-t16-new

Code Issues Pull requests Actions Projects Security and quality Insights

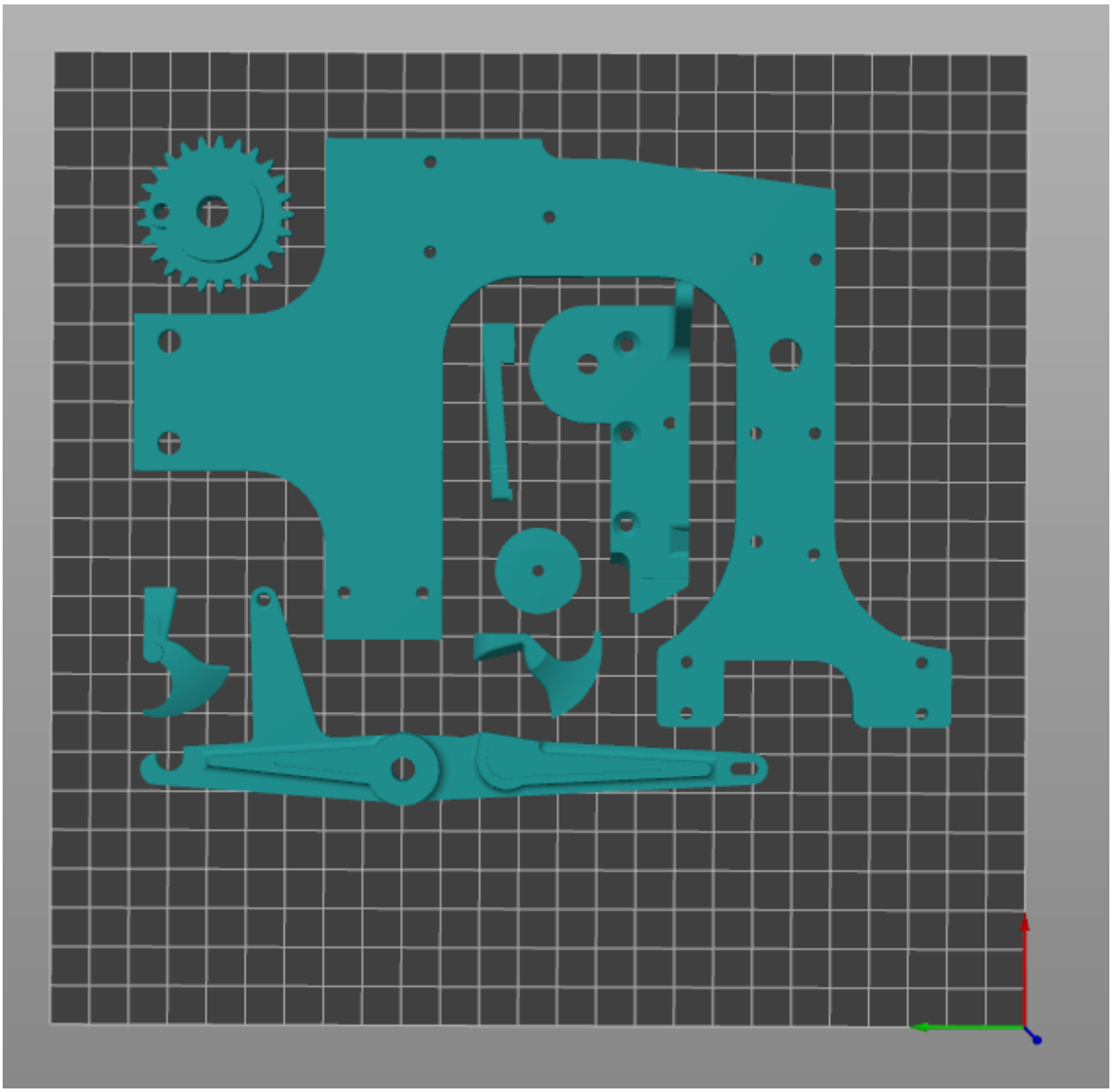
main final-project-s26-t16-new / code / touch_sensor

d-l1n Touch sensor code e15a8e6 · 3 minutes ago History

Code Blame 55 lines (41 loc) · 1.18 KB

```
#define F_CPU 16000000UL

#include <avr/io.h>
#include <util/delay.h>
#include <stdlib.h>
#include <stdio.h>
#include "uart.h"

// Initialize the ADC to read from pin PC0
void initialize() {
    // Clear power reduction bit for ADC
    PRR0 &= ~(1 << PRADC);

    // Select AVCC as reference voltage and set channel 0
    ADMUX = (1 << REFS0);

    // Disable digital input on PC0
    DIDR0 |= (1 << ADC0D);

    // Set the ADC clock to pre-scale to 128
    ADCSRA |= (1 << ADEN) | (1 << ADPS0) | (1 << ADPS1) | (1 << ADPS2);
}

// Read a value from the ADC
uint16_t adc_read(uint8_t ch) {
    // Select channel 0 (PC0)
    ADMUX = (ADMUX & 0xF8) | (ch & 0x07);

    // Start single conversion by setting ADSC
    ADCSRA |= (1 << ADSC);

    // Wait for conversion to complete
    while (ADCSRA & (1 << ADSC));

    // Return the 10-bit ADC value
    return (ADC);
```

For the physical CAD, we started from example CAD models on the internet and made the necessary adjustments to incorporate our components and detach the foot pedal from the rest of the machine to make our design more convenient and portable. Some of our components that we sent for printing are shone below.



Finally, we met as a team and aligned on direction and plans/goals for next week.

Current state of project

Currently, the project is off to a good start. We have a very basic preliminary MVP of our "foot-pedal" system (both circuit and code) which is a core part of the project that will ultimately allow the user to control the sewing machine. We also have a first version for the gears and enclosure of the sewing machine.

Right now, the biggest bottleneck is just waiting for components to come in. That is, we have ordered electronic parts and are awaiting those to ship. We have also sent out 3D print jobs to RPL and are waiting for the first print to come back so we can iterate. The reason we want to the physical enclosure to come right now 1. printing takes a long time and 2. we want to make sure the elecontric components fit within the mechanical CAD and the gear works as expected.

In the meantime we are able to progress with components that are in Detkin. Some of these are parts we will use in our final version (feather, LCD screen, Hall sensor etc.) and some are alternate versions we are just using for test (force sensor, small motor)

Next week's plan

Estimated time: 2 hours Assigned: Daniel, Chris

Finalize foot-pedal subsystem - stress test code, integrate feather for wifi communication, integrate LCD screen displaying status. Being done means have a working system that consistently measures accurate pressue, is able to send it wirelessly and display it visually. Provided 3D print finishes in time, we also want to have this integrated. This group will also be responsible for following up with the 3D print and ensuring all components are printed/accounted for.

Estimated time: 2 hours Assigned: Evelyn, Ivy

Begin the code and electronics to measure the amount of yarn left/used. This will mean building and testing the hall sensor electroncs and writing the code to support this. We also want to integrate feather for wifi communication on the ATMega connected to this system. Being done means having a hall sensor that can accurate measure rotations and communicte that via wifi.

*Note: not too much work assigned for this sprint due to midterms and team plans to meet for long work session the Sunday immediately after Sprint 2 is due.

Sprint Review #2

Last week's progress

Current state of project

Next week's plan

MVP Demo

Final Report

Don't forget to make the GitHub pages public website! If you've never made a GitHub pages website before, you can follow this webpage (though, substitute your final project repository for the GitHub username one in the quickstart guide): https://docs.github.com/en/pages/quickstart

1. Video

2. Images

3. Results

3.1 Software Requirements Specification (SRS) Results

ID	Description	Validation Outcome
SRS-01	The IMU 3-axis acceleration will be measured with 16-bit depth every 100 milliseconds +/-10 milliseconds.	Confirmed, logged output from the MCU is saved to "validation" folder in GitHub repository.

3.2 Hardware Requirements Specification (HRS) Results

ID	Description	Validation Outcome
HRS-01	A distance sensor shall be used for obstacle detection. The sensor shall detect obstacles at a maximum distance of at least 10 cm.	Confirmed, sensed obstacles up to 15cm. Video in "validation" folder, shows tape measure and logged output to terminal.

4. Conclusion

References